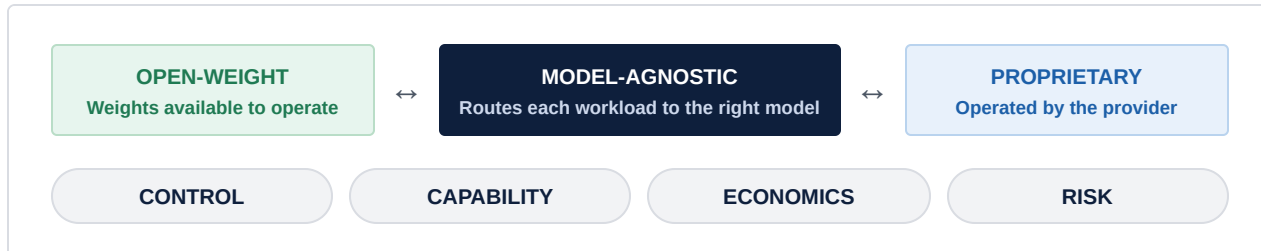


Open-Weight vs. Proprietary AI

An enterprise decision framework for control, economics, risk, and strategic autonomy

How enterprises should decide where to use proprietary models, open-weight models, and model-agnostic architectures — workload by workload, not category by category.



*The enterprise question is not which model wins.
It is where each model belongs.*

Executive Perspective: The CXO Decision Sheet

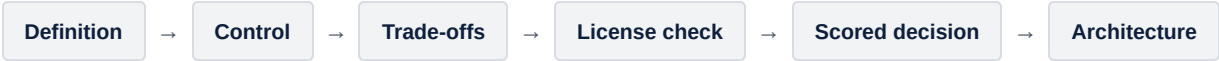
The strategic conclusion. Enterprises should stop treating AI model selection as an enterprise-wide “open vs. proprietary” decision. The right model is a **workload-level** decision, made inside a **model-agnostic architecture** capable of using both.

	Proprietary	Open-weight
Primary advantage	Frontier capability + managed operations	Control + customization
Primary trade-off	Provider dependency	Operating responsibility
Best fit	Variable demand, frontier reasoning	Sensitive, specialized or high-volume workloads where control, customization or economics justify the open-weight option
Strategic implication	Consume AI as a service	Retain the option to operate AI as an enterprise capability

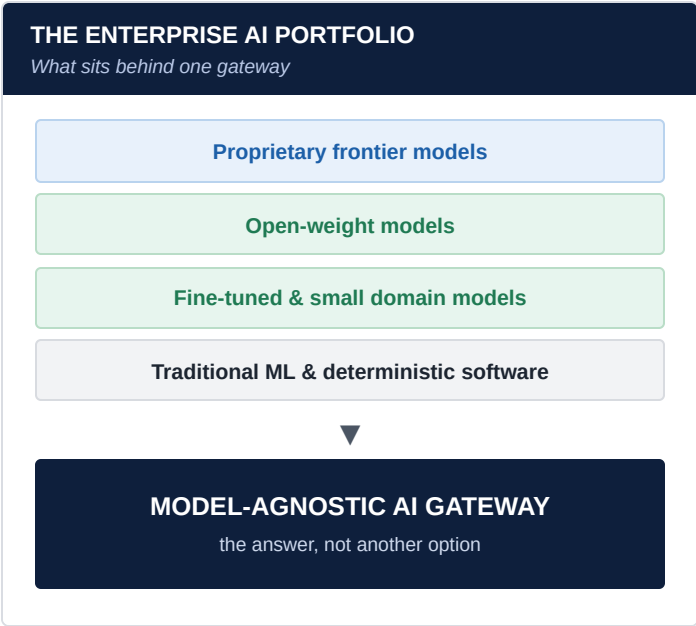
Three arguments this report makes in full: (1) “open-weight” and “open source” are not synonyms — see Chapter 1; (2) the license attached to a specific model, not its category, determines an enterprise’s real rights — see Chapter 15; (3) optionality has a price — self-hosting transfers real cost and responsibility onto the enterprise — see Chapters 6–8.

How this guide is organized

This report follows a consistent path so it can be used as a working reference, not only read start to finish: it fixes the terminology, establishes control as the strategic question, walks the eight dimensions where the two paths genuinely diverge, then turns that analysis into a licensing heatmap, a scored decision test, and a target architecture.



Part	Start here if you...
Part I	need the terminology precise, and need the “it’s about control” framing before a procurement or architecture conversation
Part II	are weighing control, data, cost, security, or governance trade-offs for a specific workload
Part III	are ready to score a workload, check a license, and stand up a target architecture



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A note on this edition

This edition is designed as an executive decision framework for evaluating proprietary, open-weight and model-agnostic AI architectures. Factual claims and source methodology are documented in Appendix A.

PART I · TERMINOLOGY & THE STRATEGIC CONTROL THESIS

Foundations

Why “open” is not one thing, and why control — not price — is the question that actually matters.

CHAPTER 1 · FRAMING

The real question is no longer “which model is better?”

Proprietary vs. open-weight looks like one line on a procurement form. It is really several separate decisions about architecture, data, cost, and control.

The AI industry has settled into two broad delivery philosophies for foundation models. **Proprietary models** keep their weights inside the provider's infrastructure and reach the enterprise through a hosted application or API. **Open-weight models** ship their trained parameters as a download, so an organization can run, adapt, and operate the model on infrastructure of its own choosing.

At first glance this looks like a simple trade between openness-and-control on one side and convenience-and-frontier-capability on the other. In practice the decision reaches much further into the enterprise than a procurement checkbox — it touches architecture, data sovereignty, security posture, regulatory exposure, customization capability, operating model, total cost of ownership, vendor dependency, model lifecycle management, and, ultimately, an organization's strategic autonomy over one of the most consequential technologies it will run this decade.

Three categories, not two

Most public discussion collapses the landscape into a binary — “open” versus “closed.” A more accurate map uses three categories, compiled directly from the Open Source Initiative's Open Source AI Definition (Oct. 2024).¹

Category	What is available	Typical deployment
Proprietary / closed	API or application access only; weights never leave the vendor	Vendor-operated cloud
Open weight	Downloadable model parameters; training code, data, and eval infrastructure may remain closed	Self-hosted, private cloud, or third-party inference platform
Open Source AI	Parameters <i>plus</i> training code <i>plus</i> sufficient training-data information, under OSI's four freedoms	Maximum transparency and reproducibility

Calling every model with downloadable weights “open source” is not just loose language — it is frequently inaccurate under any rigorous definition. OSI's companion explainer states plainly that open weights “differ significantly from Open Source AI because they do not include” the training code or data transparency needed to reproduce or audit how a model was built.² This is genuinely contested: Meta has publicly disputed OSI's characterization of the Llama license as “not open source”³; from the opposite direction, the Software Freedom Conservancy argues OSI's own definition is *too* permissive.⁴ No license from any vendor has been independently certified to satisfy every position in that debate — Ai2's OLMo family comes closest by consensus, but even that is a claim repeated by third-party trackers, not a formal certification.⁶

ENTERPRISE TAKEAWAY

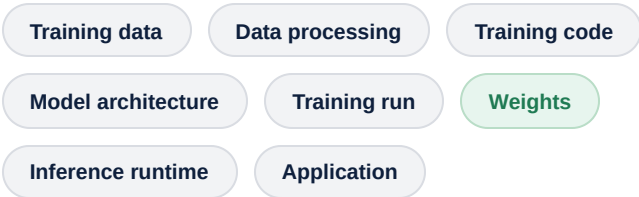
Treat open source AI, open-weight, and proprietary as three distinct categories with three distinct sets of legal rights — never as two labels for the same underlying thing.

CHAPTER 2 · FOUNDATIONS

What an AI model actually is, and where control sits

The weights are one link in a chain from training data to application, and the architectural fork between the two paths follows directly from that chain.

Before comparing the two approaches, it helps to separate an AI system into its components, because “the model” is really a pipeline:



The weights are the learned parameters produced by the training run. They are extraordinarily valuable, but they are one link in that chain, not the whole chain — which is precisely why OpenAI describes its gpt-oss family as **open-weight models**, not open-source models: released under Apache 2.0 with no field-of-use restriction, while the training data, training code, and much of OpenAI’s surrounding infrastructure remain proprietary.⁷

Proprietary model: Enterprise application → API / managed platform → model provider → model + infrastructure + runtime, all controlled by the provider. The enterprise controls the application layer and its surrounding integration, but not the underlying model weights or provider-controlled lifecycle.

Open-weight model: Enterprise application → AI gateway / inference layer → open-weight model (self-hosted, private cloud, hyperscaler, or managed inference) → enterprise-controlled or managed infrastructure. The enterprise can decide where the model runs, which inference engine and quantization scheme are used, whether and how it is fine-tuned, and when — or whether — it is upgraded — deployment optionality that Chapter 8 examines in full.

That single shift — from *consuming* a capability to *operating* one — is the architectural fork everything else in this comparison flows from.

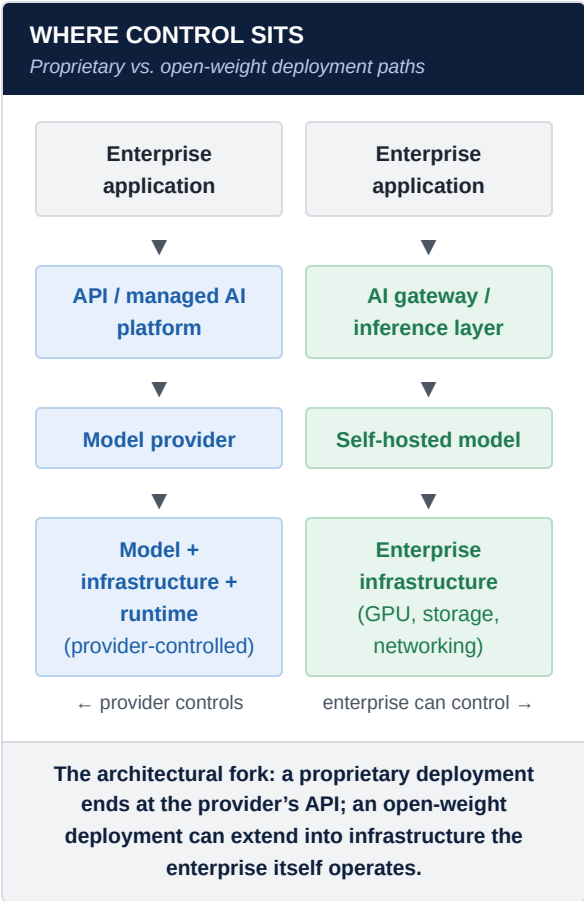


Figure 1. Where control sits: the architectural fork between proprietary and open-weight deployment paths.

The real strategic choice is control

“The strategic value of open-weight AI is not simply that it can be cheaper. It is that it changes who controls the capability.”

Proprietary AI

You consume the capability.

The provider controls:

- the weights
- the infrastructure
- the update cadence
- the deprecation & lifecycle timeline
- availability
- pricing

Open-weight AI

You can operate the capability.

The enterprise controls:

- model version
- deployment location
- inference & quantization
- customization / fine-tuning
- infrastructure
- upgrade timing

Control is not free. It becomes an operating capability.

Three separate decisions, often conflated

The debate is usually framed as one binary choice. It is actually three decisions, and an enterprise should make them explicitly and separately — the architecture chapters that follow (Part II) address each in turn:

#	Decision	The question
1	Which model?	Open-weight vs. proprietary — addressed per workload in Chapters 4–12.
2	Where does it run?	Vendor cloud, managed inference, private cloud, or on-prem — see Chapter 8.
3	Who operates it?	The vendor, a cloud provider, or the enterprise itself — see Chapter 8.

Collapsing all three into a single “open vs. closed” conversation is what produces weak architecture and governance decisions. The chapters ahead treat them as related but distinct.

THE THREE AI ARCHITECTURE DECISIONS

A signature framework, referenced throughout this report

MODEL

What capability do we use?

DEPLOYMENT

Where does it run?

OPERATING MODEL

Who carries the responsibility?

Three questions, three separate answers — a workload can be open-weight, run on a hyperscaler, and be operated by a managed-service partner, all at once.

Open-weight is a model property. Self-hosting is a deployment and operating choice.

PART II · ENTERPRISE TRADE-OFFS

Where the Two Paths Genuinely Diverge

Eight dimensions — control, data, customization, economics, operating model, lock-in, security, and governance.

CHAPTER 4 · STRATEGY

Control and strategic autonomy

Open weights can reduce dependency on a single model provider. They do not eliminate dependency — they relocate it to GPUs, clouds, and tooling.

- 7

levers a proprietary provider controls by default — weights, cadence, deprecation, infrastructure, policy, availability, pricing
- 5

levers an open-weight enterprise can control — version, location, config/fine-tuning, quantization, upgrade schedule

This is arguably the most consequential enterprise-level distinction, because it compounds over years rather than showing up in a single procurement decision. With a **proprietary model**, the enterprise consumes AI capability as a service — in the same category as any SaaS dependency, with the difference that the “service” is also the enterprise’s reasoning layer. With an **open-weight model**, the enterprise gains what is increasingly discussed as **strategic autonomy**: reduced dependence on any single provider for a core capability.

It is important to be precise about what this does and does not solve. Open weights do not eliminate vendor dependency; they relocate it. An enterprise running open-weight models can still become dependent on GPU vendors, cloud providers, inference-serving platforms, and model-specific tooling. The correct statement is narrower than the popular one: **open weights can reduce model-provider dependency, but they do not automatically eliminate technology dependency**. They trade one form of lock-in for a different, more diffuse set of them — a trade some enterprises should make and others should not.

EXECUTIVE IMPLICATION

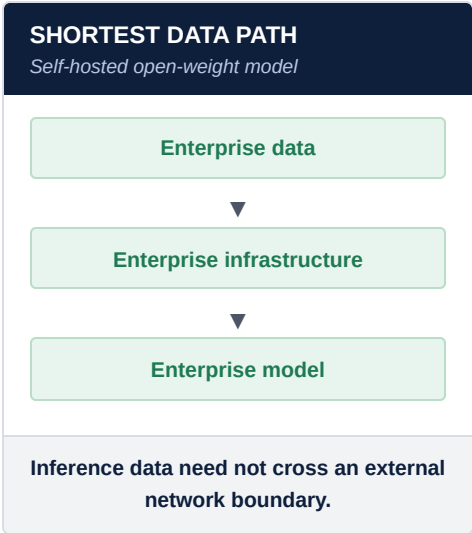
Reduced provider dependency is real, but it is not the same as reduced dependency, full stop. Budget for a new dependency surface (GPU/cloud/tooling) before claiming the win.

PRACTITIONER TAKEAWAY

Inventory which of the seven proprietary-controlled levers matter most for each workload before deciding whether open-weight autonomy is worth acquiring.

Data sovereignty and privacy

Self-hosting can shorten the data path to enterprise infrastructure → enterprise model. It does not automatically make that path secure.



Matters disproportionately for regulated industries, confidential IP, unpublished financial data, government/defense workloads, and jurisdictions with data-localization rules.

This is the dimension where open-weight models can carry a genuine, architecture-level advantage, because self-hosting changes where data physically goes. OpenAI itself lists data residency and infrastructure control as reasons an organization might choose gpt-oss over the hosted API.⁷

Proprietary vendors have, to their credit, built real controls rather than leaving this entirely to self-hosting. OpenAI does not use API/enterprise data to train by default and offers Zero Data Retention plus regional residency in the US, EU, and UAE.^{8,9} Anthropic's commercial terms state it “may not train models on Customer Content,” and it offers ZDR and US-only inference — though it does not publish a dedicated EU/regional residency product; enterprises needing that typically reach Claude via AWS Bedrock or Google Cloud Vertex AI.^{10,11,12} Google offers at-rest residency across ten countries, though that guarantee covers storage location, not necessarily where inference computation occurs.¹³

The important qualification, in both directions: self-hosting does not automatically make a system secure or compliant, and a proprietary vendor's default posture does not automatically make it non-compliant. Self-hosting moves the security boundary — and the responsibility for encryption, access control, logging, patching, and incident response — from provider to enterprise. That is a genuine transfer of both control and burden, not a free upgrade.

EXECUTIVE IMPLICATION

Data sovereignty requirements (regulatory, contractual, IP-sensitivity) are the single strongest pull toward open-weight self-hosting — but only once the security capability in Chapter 10 exists.

PRACTITIONER TAKEAWAY

Map each workload's data-boundary requirement against what proprietary ZDR / regional-residency options actually cover before assuming self-hosting is required.

Customization and performance: two different ceilings

“Open-weight” tells you almost nothing about your fine-tuning rights until you read the specific license on the specific model.

Proprietary ceiling	Open-weight ceiling
Prompting, RAG, tool-calling, hosted fine-tuning APIs within provider constraints	Full-parameter fine-tuning, LoRA, quantization, distillation, direct architectural modification

DeepSeek — the permissive example
MIT license, explicitly allows commercial use, modification, and distillation for training other LLMs.¹⁴ Its own R1-Distill variants built on Llama, however, inherit Llama’s license — provenance matters even within one vendor.

This freedom is not evenly distributed across the open-weight landscape — it depends entirely on the license attached to a given model. Alibaba’s Qwen2.5-72B permits commercial use and fine-tuning but requires a separate license once a downstream product exceeds 100M monthly active users.^{15,16} Meta’s Llama 4 Community License permits commercial fine-tuning but requires a separate license above 700M MAU and mandates “Built with Llama” branding.¹⁷ Mistral’s catalog is split model-by-model: current-generation releases ship under unrestricted Apache 2.0, while several (Mistral Large 2.1, Pixtral Large) remain under the non-commercial Mistral Research License.^{18,19,20} The full model-by-model comparison is in Chapter 15.

Performance: proprietary does not automatically mean better

Historically, proprietary frontier models held a substantial, stable capability lead. That gap has narrowed considerably, and the more useful framing is no longer “which is smarter” but “which is better matched to this workload.” Proprietary providers retain a real advantage at the **frontier capability** layer — enormous training compute, proprietary data, continuous alignment work. But there is a separate axis, **deployment capability**, where a fine-tuned, quantized, domain-constrained open-weight model can outperform a larger proprietary model on a specific enterprise task. The Linux Foundation’s Model Openness Framework was built explicitly to address models “falsely promoted” on open weights alone, when actual code/data/documentation openness varies enormously across releases with similar licensing headlines.²¹

EXECUTIVE IMPLICATION
The best general-purpose model on a public leaderboard is not necessarily the best enterprise model for a given workload.

PRACTITIONER TAKEAWAY
Evaluate on task-specific business performance measured against the enterprise’s own data and use case — not benchmark rank alone — and confirm commercial fine-tuning rights before designing around them.

Economics and total cost of ownership

Proprietary pricing is a token formula. Open-weight pricing is an infrastructure cost base — and neither one automatically beats the other.

TWO DIFFERENT COST MODELS

What actually goes into the price

Proprietary

Input tokens + Output tokens + Platform charges

\$2–\$5 / M input · \$10–\$30 / M output at the frontier high end (Claude Opus 5: \$5/\$25; Claude Sonnet 5: \$2/\$10)²²

Open-weight

GPU/accelerator + Cloud + Storage + Networking + Inference engineering + Ops + Power + MLOps + Security + FinOps

On-demand GPU capacity: \$6–\$7/GPU-hr (specialized clouds) to \$11–\$12/GPU-hr (hyperscaler)²⁴ — a fixed cost regardless of tokens served

Open-weight does not mean free: OpenAI states this explicitly for gpt-oss — weights are free, compute/storage/hosting are not.⁷

WHERE THE ECONOMICS CAN CROSS OVER

Cost per request vs. volume / utilization

There is no universal breakeven point. The crossover depends on utilization, hardware, model size, latency requirements, engineering cost, and workload stability.

Volume profile	Likely advantage
Low / experimental	Proprietary
Highly variable	Proprietary
High & predictable, well-utilized	Open-weight (potential)
High customization need	Open-weight (strategic)

Practitioner reports on the vLLM serving framework's community forum describe peak GPU utilization of 70–90% achievable with tuning, but **average utilization commonly landing at only 40–70% in production**.²⁵ A broader VentureBeat audit citing Cast AI fleet data found average enterprise GPU utilization across all workloads as low as roughly 5% (illustrative rather than LLM-specific).²⁶ A rigorous, independently published “breakeven token volume” does not appear to exist in credible public research as of this writing; widely circulated figures are not methodologically transparent enough to cite.

EXECUTIVE IMPLICATION

The token-price comparison sheet finance teams usually build is incomplete without the nine-layer operating stack (plus FinOps) in Chapter 8 priced alongside it.

PRACTITIONER TAKEAWAY

Model TCO using measured production utilization — practitioner evidence commonly reports 40–70%, well below sticker GPU-hour rates — before comparing to proprietary token pricing.

Operating model: three decisions, not one

Self-hosting is not a model choice. It is an operating-model commitment — and the question of who operates the capability deserves its own explicit answer.

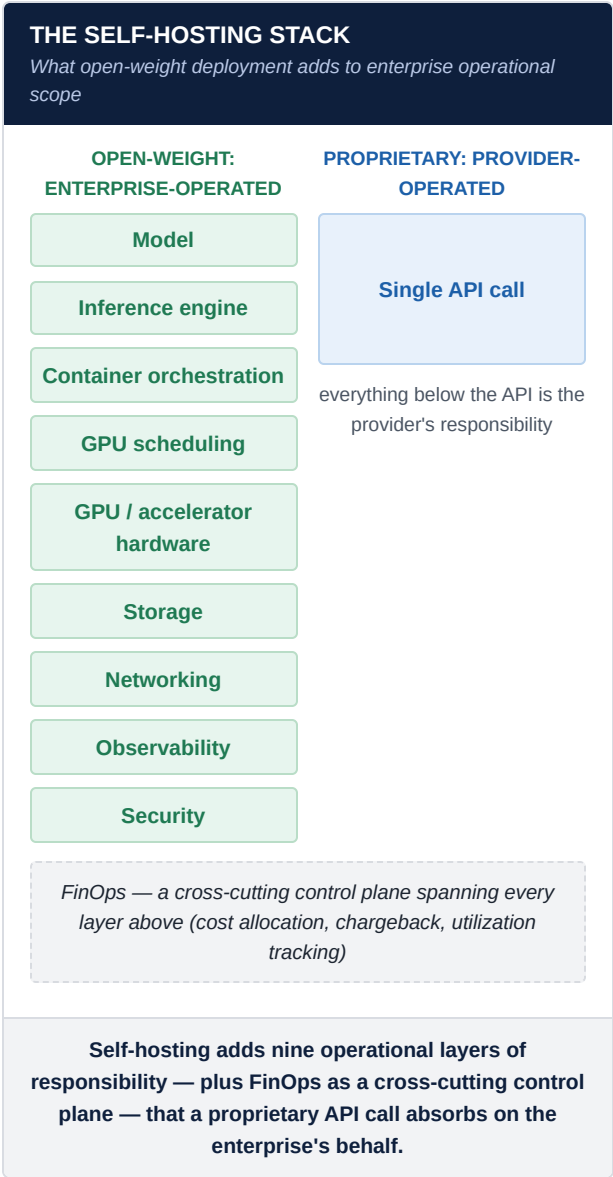


Figure 2. The self-hosting stack: what open-weight deployment adds to enterprise operational scope.

EXECUTIVE IMPLICATION

Self-hosting is a nine-layer operating-model commitment plus a FinOps control plane across all of them — and average GPU utilization in production commonly lands at 40–70%, not the 90%+ a sticker-price estimate assumes.

PRACTITIONER TAKEAWAY

Don't approve open-weight self-hosting without a named owner for each of the nine operational layers, plus a FinOps owner — an unowned layer becomes an unmanaged risk, not a savings.

Fundamentally: under a proprietary model, **AI is consumed as a service**. Under an open-weight model, **AI can become an enterprise-controlled technology asset** — provided the enterprise is willing and able to operate it as one. This does not mean every enterprise should self-host. It means the open-weight path keeps that choice available, while the proprietary path, by design, does not.

Self-hosting does not mean choosing a different model — it means taking on a different **operating model** for AI. Every layer in the stack at left is a real engineering and operational commitment, and the gap between theoretical and realized efficiency is larger than most procurement models assume.

This is a strategic decision disguised as a technical one. An organization choosing open-weight self-hosting is implicitly choosing to build or buy an AI infrastructure operations capability — a capability that, once built, becomes reusable across future models and workloads, but that carries real upfront and ongoing cost regardless of how favorably the per-token math looks on a vendor's pricing page.

The three decisions, restated

1. Which model? Open vs. proprietary. **2. Where does it run?** Vendor cloud / managed inference / private cloud / on-prem. **3. Who operates it?** Vendor, cloud provider, or enterprise. Answer all three explicitly — per workload.

Vendor lock-in and model lifecycle management

Neither path avoids vendor lock-in, and neither hands off lifecycle responsibility for free — each just relocates both to a different layer of the stack.

Under proprietary	Lock-in accumulates at
API contract, model behavior/prompt sensitivity, SDK, tool-calling interface, embeddings tuned to provider representations, safety/moderation policy baked into application logic.	
Under open-weight	Lock-in shifts to
Cloud/GPU provider, inference engine and its optimizations, MLOps/orchestration platform, proprietary tooling built around deployment.	

Accurate summary: proprietary models concentrate dependency at the provider layer; open-weight models lower model-provider dependency while raising operational and infrastructure dependency.

A system deeply optimized around one provider’s model behavior rarely migrates cleanly to a different provider’s model, even when the new model is nominally more capable, because so much invisible tuning — prompt engineering, few-shot examples, guardrail logic — was built against the quirks of the original model.

Model lifecycle management

Under a **proprietary** model, the provider manages the full lifecycle, and leading providers now publish formal deprecation commitments: OpenAI commits to a minimum of six months’ notice before deprecating a GA model (as little as two weeks for preview models).³¹ Anthropic commits to at least 60 days’ notice and has separately committed to preserving the weights of every publicly released model “for, at minimum, the lifetime of Anthropic as a company.”^{32,33} Google publishes shutdown dates but does not guarantee a specific minimum notice window.³⁴ These are genuine, checkable commitments that de-risk the “the provider pulls the model out from under us” scenario.

Under an **open-weight** model, the enterprise itself manages selection, validation, monitoring, patching, versioning, and retirement — requiring a model registry, evaluation/regression testing before upgrades, artifact security scanning, drift monitoring, and a rollback plan. An enterprise that adopts open-weight models without this capability tends to end up worse off than a proprietary customer — running an unpatched, unmonitored model for years past the point a comparable proprietary deployment would have received an automatic security update.

EXECUTIVE IMPLICATION

Contractual deprecation-notice commitments have narrowed the lifecycle gap, but they do not close it — the two paths still concentrate risk at different layers.

PRACTITIONER TAKEAWAY

Stand up a model registry and rollback plan before adopting open-weight models in production, or the control gained is illusory.

Security: a counterintuitive trade-off

Open-weight deployment increases control, and control increases responsibility. Neither category is categorically more secure.

It would be a mistake to claim either “open models are more secure” or “proprietary models are more secure.” NIST’s AI Risk Management Framework Generative AI Profile (NIST AI 600-1) explicitly names “levels of model access or availability of model weights” as one variable shaping a system’s risk profile — alongside architecture, training data, and use-case context — rather than treating weight availability as inherently safer or riskier.²⁷

Proprietary		Open-weight	
Advantages	Risks	Advantages	Risks
Hyperscale-investment managed infrastructure; centralized continuous patching; provider-scale monitoring & IR; formal certifications (SOC 2, ISO 27001, HIPAA BAAs)	Reduced visibility into model internals; provider becomes part of trust boundary; dependency on provider’s security posture; limited ability to verify a specific deployed version	Runs inside enterprise’s own security boundary; deeper inspectability; independent red-teaming on own schedule; full control of patch timing/rollback	Malicious/compromised model artifacts entering the supply chain; insecure serving infra without deep ML security experience; model extraction/exfiltration risk; full patching/monitoring responsibility

The NTIA’s 2024 report on dual-use foundation models with widely available weights reached a similarly balanced conclusion after formal risk review: “current evidence is not sufficient to definitively determine either that restrictions on such open weight models are warranted, or that restrictions will never be appropriate in the future” — recommending ongoing monitoring rather than blanket restriction or blanket assurance.²⁸ (Published under the prior administration’s executive-order framework; confirm current status before treating it as live federal policy.)

Control increases responsibility.

EXECUTIVE IMPLICATION

An enterprise that self-hosts without building a security operations capability has not reduced its risk relative to a proprietary API — it has likely increased it, while believing the opposite because “we control it now” feels safer than it necessarily is.

PRACTITIONER TAKEAWAY

Budget for a dedicated ML security operations capability (artifact scanning, red-teaming, patch/rollback pipeline) as a prerequisite to self-hosting, not an optional add-on.

Governance, compliance and explainability

With proprietary models, governance questions are answered by reading a vendor’s terms. With open weights, the enterprise has to answer them itself — and access to the weights is not the same claim as being able to explain the model’s behavior.

Vendor-governance questions (proprietary)
Answered by reading the provider's terms and trusting its commitments.

Enterprise-governance questions (open-weight)
The organization itself must document, evaluate, and defend its answers — there is no vendor to point to.

Both categories must answer: where does the model run, what data does it process, who owns it, what license governs use, can it be modified/redistributed/fine-tuned commercially, what training-data information exists, can it be independently evaluated, can a version be retained indefinitely, and what happens when the provider changes the model out from under the enterprise?

The EU AI Act gives this real legal weight. Article 53(1) requires providers of general-purpose AI models to maintain technical documentation, provide integration information, maintain a copyright-compliance policy, and publish a training-content summary.²⁹ Article 53(2) exempts models “released under a free and open-source licence” (access, usage, modification, distribution of parameters, architecture, and usage information all public) from the technical-documentation and downstream-information obligations — but *not* from the copyright-policy or training-data-summary requirements, both of which still apply.²⁹ Crucially, that exemption disappears entirely once a model is designated as carrying “systemic risk” — presumed automatically above 10²⁵ floating-point operations of training compute, a threshold the European Commission describes as currently capturing “the most advanced models.”³⁰ An open license earns a real, legally defined compliance advantage under EU law today — but it is conditional and compute-bounded, not unconditional.

Explainability and transparency

Neither category automatically delivers interpretability. Having access to the weights does not mean an enterprise understands *why* a model produced a particular output. What open weights do enable, which a closed API cannot, is deeper *technical* examination: architecture inspection, activation analysis, controlled experimentation, and reproducibility studies. “We can see the weights” and “we can explain the model's behavior” are two different, only loosely related claims.

EXECUTIVE IMPLICATION
Governance obligations are increasingly written into binding law, not internal policy — the EU AI Act’s open-source exemption is real but disappears at systemic-risk scale.

PRACTITIONER TAKEAWAY
Build one governance questionnaire that both proprietary and open-weight workloads must answer — don’t let the model category decide which questions get asked.

CHAPTER 12 · FRAMING

What “open-weight” does not mean

The single most misused phrase in enterprise AI strategy. Read this page before it appears unqualified in a board deck.

NOT AUTOMATICALLY

- ✗ Open source
- ✗ Free
- ✗ Secure by default
- ✗ No vendor dependency
- ✗ Better economics
- ✗ Better performance
- ✗ Automatically compliant
- ✗ Explainable AI

OPEN-WEIGHT CAN PROVIDE

- ✓ Model version control
- ✓ Deployment control
- ✓ Fine-tuning flexibility
- ✓ Greater data-boundary control
- ✓ Greater infrastructure optionality
- ✓ Strategic autonomy from a single provider

Every item on the left has been asserted, at some point, as an automatic property of open-weight models somewhere in industry commentary. None of them are automatic — each depends on the specific license (Chapter 15), the enterprise's own security operations capability (Chapter 10), and the operating model it is willing to build (Chapter 8). The items on the right are genuinely available — but only to an enterprise willing to operate the capability, not merely download it.

ENTERPRISE TAKEAWAY

Every claim about what open-weight models provide should be qualified with “*if the enterprise builds the corresponding capability.*” Downloadable weights are a starting condition, not an outcome.

PART III · MAKING THE DECISION

From Analysis to Action

Turning the preceding trade-offs into a gated, weighted test, a workload matrix, a license heatmap, and a target architecture.

CHAPTER 13 · STRATEGY

The enterprise model selection test

Seven diagnostic questions replace the false binary of “open or proprietary” — but not all seven carry equal weight. Some are hard constraints; the rest are trade-offs to be scored.

STEP 1 · HARD GATES — ask before scoring anything

A hard gate is not a trade-off to be weighed against frontier capability or cost — a “yes” eliminates the non-compliant option outright, regardless of how well it scores elsewhere.

Gate question	If yes...
Must data remain within a specific controlled boundary (regulatory, contractual, IP)?	Eliminate any option that cannot guarantee that boundary (Ch. 5)
Is the workload's commercial use restricted by the specific model's license?	Eliminate that model regardless of capability fit (Ch. 15)
Is long-term version pinning mandatory (regulated/safety-critical)?	Eliminate options the vendor can force-upgrade without consent (Ch. 9)
Is self-hosting operationally impermissible (no infra, no security capability)?	Eliminate self-hosted open-weight deployment for this workload — managed-inference open-weight, or proprietary, remain options (Ch. 8, 10)

STEP 2 · WEIGHTED SCORING — for whatever survives the gates

Dimension	1 = low	5 = high	Weight / gate status
1. Data sovereignty	External processing acceptable	Must remain inside a controlled boundary	Gated in Step 1
2. Customization	Prompting / RAG sufficient	Deep fine-tuning required	Medium–High
3. Frontier capability	Task-specific capability sufficient	Frontier reasoning required	Medium–High
4. Infrastructure readiness	No internal AI-ops capability yet	Existing GPU / MLOps capability	Gated in Step 1
5. Inference volume	Variable / low	High / predictable, well-utilized	Medium
6. Provider dependency	Single-provider dependency acceptable	Strategically unacceptable	Low–Medium
7. Version stability	Frequent upgrades acceptable	Version must remain pinned	Gated in Step 1

Shaded rows are resolved in Step 1 and shown here only for reference — a workload's specific circumstances rarely turn a hard gate into a matter of degree.

STEP 3 · ARCHITECTURE DECISION

Proprietary. Passed all gates; low/medium scores; frontier capability or low/variable volume dominate.

Open-weight. Passed all gates; high scores on sovereignty, customization, or version stability.

Hybrid. A gate forces one component (e.g. data boundary) while another sub-task doesn't need it.

Model-agnostic. Scores split across workloads — the common case. Route by workload.

ENTERPRISE TAKEAWAY

Gate first, score second: a shaded row scored like the rest is how a hard requirement gets silently overridden. Run per workload, not once for the whole enterprise.

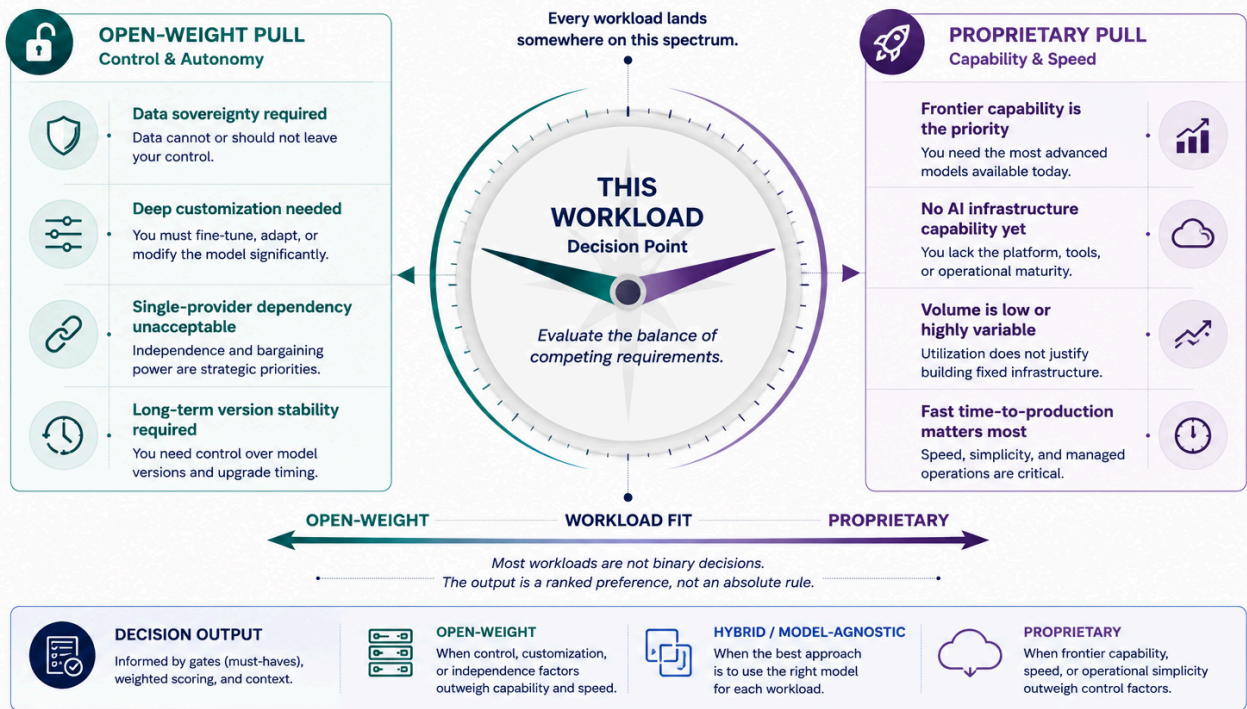
Workload decision matrix

Recommendation is workload-specific, not enterprise-wide. These six patterns are illustrations, not universal rules — see Chapter 13's test for the actual scoring.

Enterprise workload	Recommended default	Why
Customer-facing support assistant, high volume, no confidential data	Proprietary	Frontier reasoning + conversational polish; variable demand doesn't justify dedicated infrastructure
Contract / legal-document analysis on unpublished, confidential material	Open-weight (or controlled proprietary)	Data-boundary control dominates; requires the security capability from Chapter 10
High-volume, low-complexity classification / extraction	Open-weight	Unit economics + specialization win once volume justifies the fixed infrastructure cost
Domain-specialized internal tool (engineering standards, product docs)	Open-weight	Depth of customization, not raw capability, is the differentiator
Complex, open-ended reasoning task	Proprietary frontier	Raw capability dominates; deployment flexibility is secondary
Regulated, deterministic workflow requiring version pinning	Hybrid	Governance and version stability requirements dominate over either default

Figure 3 – Decision Compass

Evaluate each workload across competing requirements to determine where it should run.



IMPORTANT: This compass is workload-specific, not enterprise-wide. Reassess as requirements, volumes, risks, and model options evolve.

CHAPTER 15 · GOVERNANCE

Licensing reality check: what enterprises actually get, model by model

“Open-weight” spans permissively licensed releases and research-only or otherwise conditionally licensed releases — sometimes inside one vendor’s own catalog. Read the specific license attached to the specific model.

Model family & license	Commercial use	Fine-tune / distill	Redistribute	Key restriction	Level
OpenAI gpt-oss (120b, 20b) Apache 2.0 · OSI license ✓ ^{7,8}	✓	✓	✓	Non-binding usage policy only	●
DeepSeek R1 / V3-0324+ MIT · OSI license ✓ ¹⁴	✓	✓	✓	None of substance; R1-Distill inherits Llama's license	●
Alibaba Qwen2.5 (≤32B) / selected Qwen3 releases Apache 2.0 · OSI license ✓ ¹⁶	✓	✓	✓	None	●
Mistral 3 / Large 3 and other Apache-licensed releases Apache 2.0 · OSI license ✓ ^{18,19}	✓	✓	✓	None	●
Google Gemma 4 (Apr. 2026) Apache 2.0 · OSI license ✓ ³⁷	✓	✓	✓	Parallel Prohibited Use Policy (ToS, not license condition)	●
Meta Llama 4 Llama 4 Community License · license not OSI-approved ^{3,17}	△	△	△	Discretionary Meta approval required above 700M MAU; “Built with Llama” branding	●
Alibaba Qwen2.5-72B Qwen License Agreement · license not OSI-approved ¹⁵	△	△	△	Discretionary Alibaba license required above 100M MAU; attribution	●
Google Gemma 2 / 3 Gemma Terms of Use · license not OSI-approved ³⁶	✓	△	△	Prohibited Use Policy (content/safety/professional-practice)	●
Mistral Large 2.1 / Pixtral Large Mistral Research License · license not OSI-approved ¹⁸	✗	△	✗	Field-of-use bar on any commercial use; research only	●

✓ Allowed △ Conditional (see restriction column) ✗ Not allowed / requires a separate license. “OSI license ✓” marks the named license itself as OSI-approved — it does not certify the model as Open Source AI under OSI's fuller definition (Chapter 1). Compiled from each vendor's official license text/file, current as of August 2026. Citations 3, 7–8, 14–20, 36–37 in References; the complete clause-level comparison (MAU thresholds, attribution wording) is preserved from the original edition in Appendix A. **This is a strategic and architectural comparison, not legal advice** — verify current license text before a procurement or production decision. ● Low restriction ● Medium ● High

Two patterns are worth naming explicitly. First, **licensing has been trending more permissive across the industry** — Mistral's flagship line, Alibaba's smaller Qwen sizes, and Google's newest Gemma generation have all moved to plain Apache 2.0 over the past two years, while DeepSeek moved from a custom restrictive license to MIT within months of its initial release. Second, **the restrictive terms that remain cluster around scale, not around the base right to use the model** — the MAU thresholds in Meta's and Alibaba's licenses exist to prevent a handful of the largest technology companies from redistributing weights inside a rival consumer product, not to restrict a typical enterprise.

ENTERPRISE TAKEAWAY

Proprietary vendor terms follow a different logic entirely: an enterprise contracts for a service. OpenAI, Anthropic, and Google Cloud all publish current SOC 2 Type II and ISO/IEC 27001 certifications, with HIPAA-eligible configurations available.^{35,38} The category label “open-weight” spans permissively licensed releases and research-only or otherwise conditionally licensed releases — read the specific license attached to the specific model.

A model-agnostic architecture for large enterprises

The strongest position is not choosing a philosophy. It is a gateway that assigns the model to the workload, not the workload to the model.

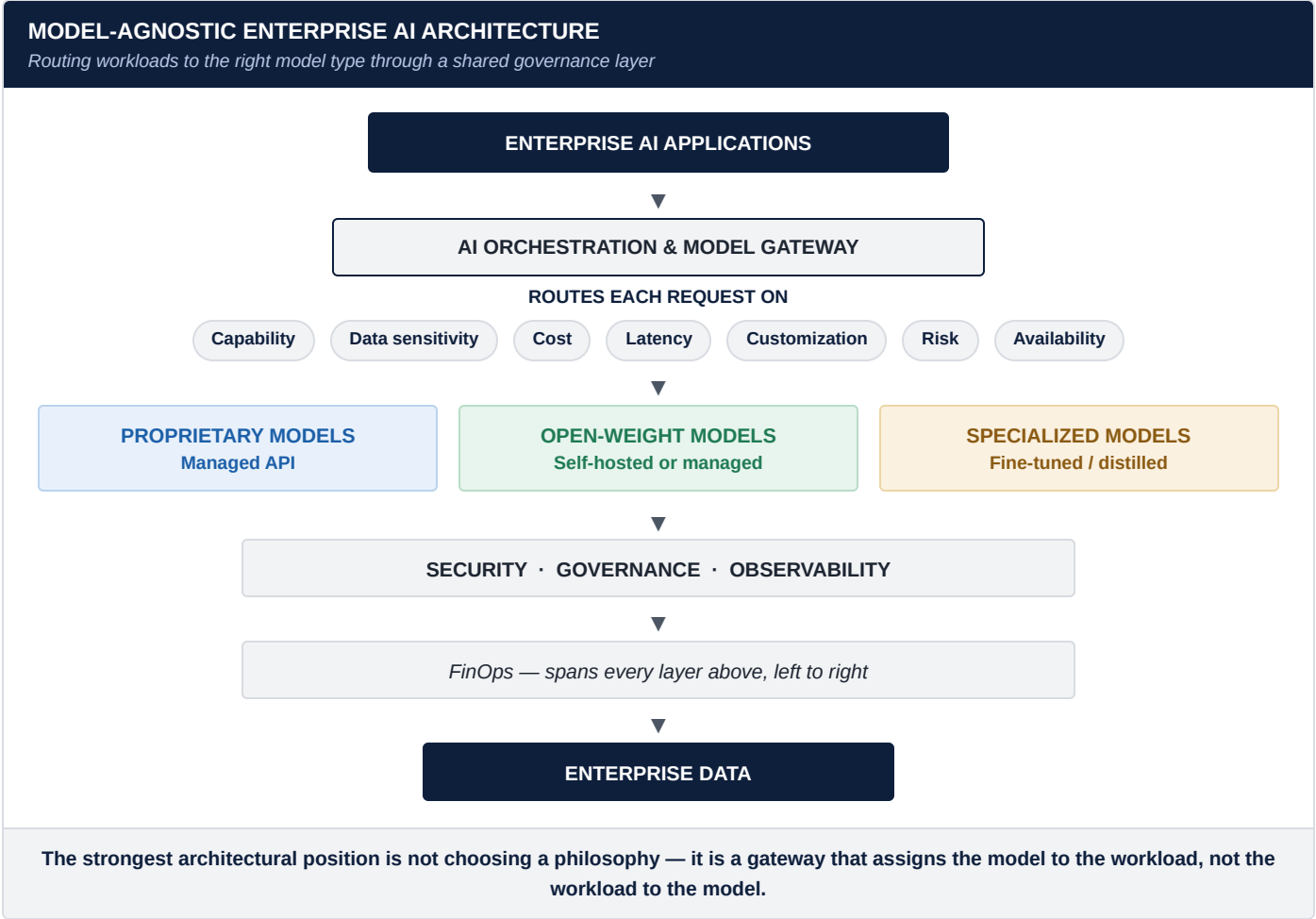


Figure 4. A model-agnostic enterprise AI architecture: routing workloads to proprietary, open-weight, or specialized models through a shared governance, security, observability, and FinOps layer.

In practice this lets an organization assign the model to the workload rather than the workload to the model: a customer-facing reasoning task might route to a frontier proprietary model; a workload touching highly confidential internal documents might route to a self-hosted open-weight model; high-volume classification might route to a small, cheap open-weight model optimized for throughput; a specialized domain task might route to a fine-tuned model; an experimental or low-volume workload might simply use a managed API rather than justify dedicated infrastructure. This is considerably more resilient than committing the enterprise's AI strategy to a single provider's roadmap, pricing, and policy decisions.

The CXO cheat sheet

If your priority is... — the single most shareable page in this report.

STRATEGY Strategic independence → Open-weight + model-agnostic Enterprise resilience → Multi-model architecture Multi-year platform bet → Model-agnostic gateway (Ch. 16)	ECONOMICS Low, variable volume → Proprietary High, predictable, well-utilized → Evaluate open-weight Fast time-to-market → Proprietary / managed inference
RISK Data sovereignty required → Open-weight / controlled deployment Long-term version pinning → Open-weight Approaching EU systemic-risk scale → Reassess Article 53(2) exemption (Ch. 11)	EXECUTION Frontier capability required → Proprietary Deep customization required → Open-weight No AI-ops capability yet → Proprietary / managed inference

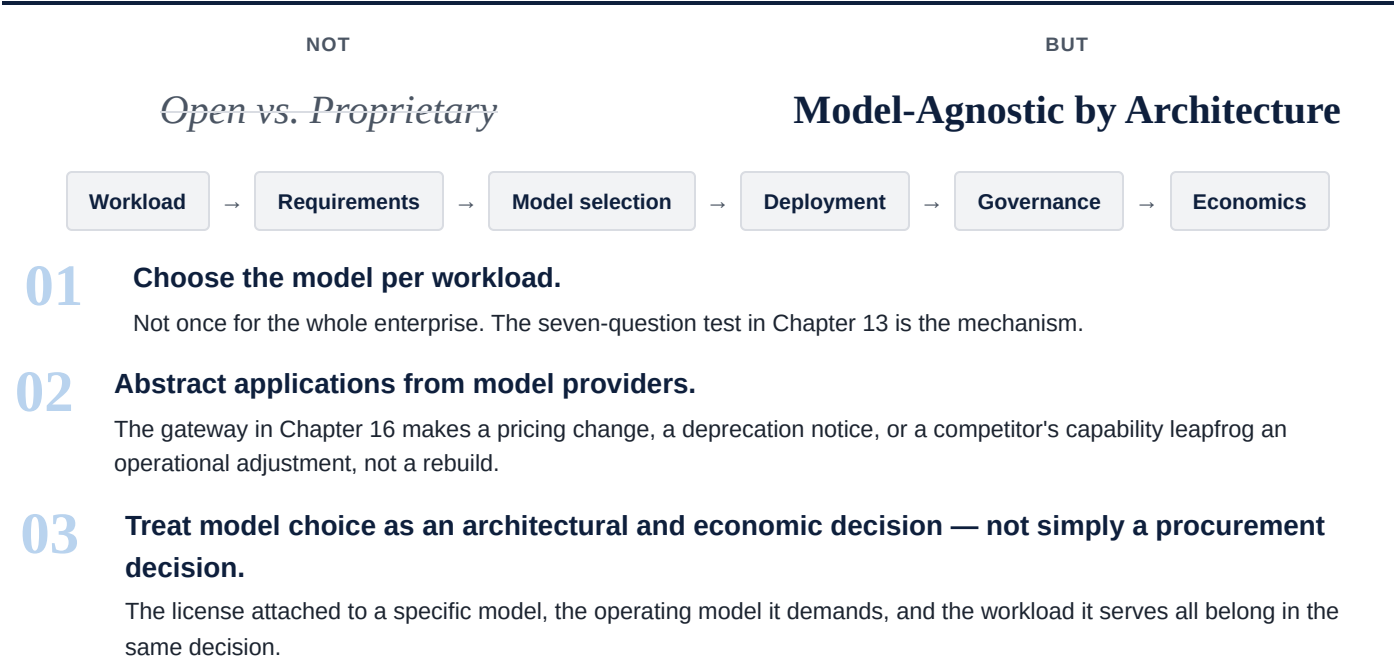
Strategic target state: MODEL-AGNOSTIC.

This page summarizes the gated, weighted test in Chapter 13 and the workload matrix in Chapter 14 — use it for a first pass, then confirm with the full test (hard gates first) before committing infrastructure spend to any single workload.

CLOSING PERSPECTIVE

CHAPTER 18 · FRAMING

The enterprise AI end state



The debate should not resolve into “open versus closed.” The strategic direction is toward treating “open, proprietary, and model-agnostic” as complementary options, not a single doctrine every enterprise adopts overnight — a mature AI stack will typically mix proprietary frontier models, open-weight reasoning models, fine-tuned domain models, and increasingly autonomous agents built across all of them.

The strategic value of open-weight AI is not that a model is cheaper or merely downloadable. Its real value is that it changes *who controls the AI capability*.

BOTTOM LINE

The mature enterprise position is not “we use open models” or “we use proprietary models.” It is an architecture capable of using both, with the model assigned per workload based on data sensitivity, customization need, economics, and governance — not a single enterprise-wide bet on either category.

APPENDICES & REFERENCE

Appendix A

Illustrative scenarios, sourcing methodology, and a glossary of eight terms worth pinning down.

Four illustrative enterprise scenarios

How the seven-question test in Chapter 13 resolves differently depending on which requirement dominates.

Customer-facing support, high volume, no confidential data
Frontier polish + variable demand → managed proprietary API is the default; revisit if volume changes the unit economics.

Contract / legal-document analysis, confidential material
Data sovereignty dominates → self-hosted open-weight, or a proprietary vendor's regional/ZDR controls if those meet the compliance bar. Requires the Ch.10 security capability.

High-volume, low-complexity classification / extraction
Cost-per-request dominates → a small, quantized, fine-tuned open-weight model typically wins on unit economics once volume justifies fixed infrastructure.

Domain-specialized internal tool
Depth of customization, not raw capability → open-weight model with unambiguous commercial fine-tuning rights, verified against the specific license.

None of these is a universal rule; they illustrate how the test resolves depending on which requirement dominates.

A note on sourcing and methodology

Every factual claim in this report — license terms, vendor policy commitments, regulatory text, and pricing — is drawn from primary sources: official license files and model cards, vendor documentation and trust portals, OSI's published definitions, the official text of Regulation (EU) 2024/1689 and NIST AI 600-1, and vendors' own pricing pages. Secondary commentary is cited only where it is the primary source for a claim about opinion or debate. Two claims commonly repeated in industry commentary were checked and could not be verified from any primary source, and are deliberately omitted: a specific "breakeven token volume" for self-hosting, and a formal OSI certification registry rating named models. Pricing, licensing, and regulatory thresholds reflect official sources as of August 2026 and should be re-verified before a procurement or compliance decision.

DISCLAIMER

This report is for strategic and architectural evaluation. It does not constitute legal advice. Licensing, contractual, and regulatory terms should be reviewed against the specific model release and applicable use case, with counsel where warranted, before a production or procurement decision.

Glossary

Open-weight model — trained parameters published for download, independent of whether training code/data/eval infrastructure are also released. **Open Source AI** — OSI's four freedoms plus training code and sufficient training-data information; parameters alone do not qualify. **Proprietary model** — weights remain under the provider's exclusive control, accessed via hosted API. **Field-of-use restriction** — a license condition limiting how/by whom a model may be used. **General-purpose AI (GPAI) model** — the EU AI Act's term for a foundation model performing a wide range of tasks. **Systemic risk (EU AI Act)** — presumed above 10²⁵ FLOPs of training compute; removes the Article 53(2) exemption. **Zero Data Retention (ZDR)** — a vendor configuration under which prompts/responses are not stored after the response. **Total cost of ownership (TCO)** — for self-hosting, the full GPU/cloud/storage/network/engineering/ops cost base.

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